工程數學 Final Exam January 12, 2021

ID:_____ Name:_____

滿分 20 分。答案若是有 ∑符號，可以不用展開，作答方式如上課所示即可

[3%] (1) 已知 F(s) 求出 f(t)

$F(s) = \frac{2e^{5s}}{s^2-2s+1}$

[3%] (2) 已知 F(s) 求出 f(t)

$L^{-1}\left\{\frac{-e^{-s}}{s(s+1)}\right\}$

[4%] (3) 求出 f(x) 的傅立葉級數

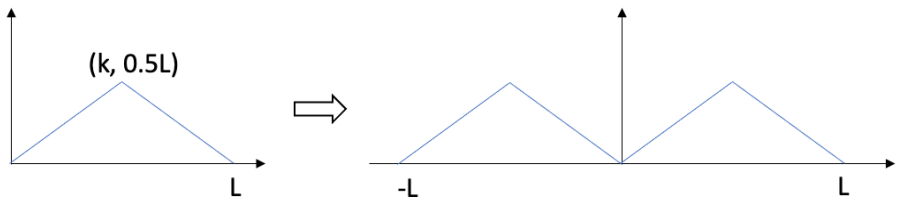
$f(x) = \begin{cases} 1, & -1<x<0 \\ x, & 0<x<1 \end{cases}$ (週期為 2)

[3%] (4) 求出 f(x) 的傅立葉級數

$f(x) = x^2 \quad , \quad -1 < x < 1$

[3%] (5)求$f(x) = 3x + 5\pi$ *當* $-\pi < x < \pi$ *且* $f(x+2\pi) = f(x)$*之傅立葉級數*

[4%] (6) [半幅展開式] 求 $f(x) = \begin{cases} \frac{2k}{L}x & if\ 0 < x < \frac{L}{2} \\ \frac{2k}{L}(L-x) & if\ \frac{L}{2} < x < L \end{cases}$ 的偶函數延伸



公式表:

	$f(t)$	$\mathcal{L}(f)$
1	1	$1/s$
2	t	$1/s^2$
3	t^2	$2!/s^3$
4	t^n $(n = 0, 1, \cdots)$	$\frac{n!}{s^{n+1}}$
5	t^a (a positive)	$\frac{\Gamma(a+1)}{s^{a+1}}$
6	e^{at}	$\frac{1}{s-a}$
7	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
8	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
9	$\cosh at$	$\frac{s}{s^2-a^2}$
10	$\sinh at$	$\frac{a}{s^2-a^2}$
11	$e^{at}\cos \omega t$	$\frac{s-a}{(s-a)^2+\omega^2}$
12	$e^{at}\sin \omega t$	$\frac{\omega}{(s-a)^2+\omega^2}$

First Shifting Theorem, s-Shifting

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

$$e^{at}f(t) = \mathcal{L}^{-1}\{F(s-a)\}$$

Laplace Transform of Derivatives

$$\mathcal{L}(f') = s\mathcal{L}(f) - f(0)$$

$$\mathcal{L}(f'') = s^2\mathcal{L}(f) - sf(0) - f'(0)$$

$$\mathcal{L}(f^{(n)}) = s^n\mathcal{L}(f) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$$

Laplace Transform of Integral 積分轉換

$$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{1}{s}F(s), \qquad \text{thus} \qquad \int_0^t f(\tau)\,d\tau = \mathcal{L}^{-1}\left\{\frac{1}{s}F(s)\right\}$$

Second Shifting Theorem; Time Shifting

$$\tilde{f}(t) = f(t-a)u(t-a) = \begin{cases} 0 & \text{if } t < a \\ f(t-a) & \text{if } t > a \end{cases}$$

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

$$f(t-a)u(t-a) = \mathcal{L}^{-1}\{e^{-as}F(s)\}$$

Examples for t-shifting by 1

$1 \mapsto \frac{1}{s}$ $\qquad u(t-1) \mapsto e^{-s}\frac{1}{s}$

$t \mapsto \frac{1}{s^2}$ $\qquad (t-1)u(t-1) \mapsto e^{-s}\frac{1}{s^2}$

$t^2 \mapsto \frac{2}{s^3}$ $\qquad (t-1)^2u(t-1) \mapsto e^{-s}\frac{2}{s^3}$

$\sin t \mapsto \frac{1}{s^2+1}$ $\qquad \sin(t-1)u(t-1) \mapsto e^{-s}\frac{1}{s^2+1}$

傅立葉級數 ***(週期 2L)***

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{L} x + b_n \sin \frac{n\pi}{L} x \right)$$

(a) $$a_0 = \frac{1}{2L} \int_{-L}^{L} f(x)\, dx$$

(b) $$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos \frac{n\pi x}{L}\, dx \qquad n = 1, 2, \cdots$$

(c) $$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin \frac{n\pi x}{L}\, dx \qquad n = 1, 2, \cdots$$

Fourier <u>Cosine</u> Series (*even function, 偶函數*) *(週期 2L)*

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{L} x \qquad (f \text{ even})$$

$$a_0 = \frac{1}{L} \int_{0}^{L} f(x)\, dx, \qquad a_n = \frac{2}{L} \int_{0}^{L} f(x) \cos \frac{n\pi x}{L}\, dx, \qquad n = 1, 2, \cdots$$

Fourier <u>Sine</u> Series (*odd function, 奇函數*) *(週期 2L)*

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{L} x \qquad (f \text{ odd})$$

$$b_n = \frac{2}{L} \int_{0}^{L} f(x) \sin \frac{n\pi x}{L}\, dx$$